



AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1046 - OO Analysis & Design

PROGRAMME(S):

CASE	BSc Computer Apps (Soft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

YEAR OF STUDY: 1,3

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 20]****Q 1(a)****[5 Marks]**

Read the description in the Appendix of the VIP Cars Hire system and develop a Use Case diagram showing the main Actors and Use Cases.

Q 1(b)**[15 Marks]**

Write the text for three of your chosen Use Cases using the Cockburn template and include some variations and extensions.

[End of Question 1]**QUESTION 2****[TOTAL MARKS: 20]****Q 2(a)****[12 Marks]**

Develop a State Diagram for the RESERVATION object based on the Appendix. Make sure to detail the guards on every transition.

Q 2(b)**[4 Marks]**

Suggest a sub-state that could be created in your model and show what it would contain.

Q 2(c)**[4 Marks]**

Explain the purpose and give an example of the use of the History transition link of State Diagrams.

[End of Question 2]

QUESTION 3

[TOTAL MARKS: 20]

Q 3(a)

[14 Marks]

Add four more classes to the existing RESERVATION and create a Class Diagram for VIPCars. Include role names, inheritance, multiplicity and association classes in your diagram. Have at least two variables and two functions in each.

Q 3(b)

[6 Marks]

Explain *dependency* among classes. Give an example of three types.

[End of Question 3]

QUESTION 4

[TOTAL MARKS: 20]

Q 4(a)

[10 Marks]

Use the Object Constraint Language to write an invariant for the RESERVATION class, in the style:

context <classifier>

inv [<constraint name>]: <Boolean OCL expression>

Q 4(b)

[5 Marks]

Write a precondition for each of two functions that you have identified in any of your classes.

context <classifier>::<operation> (<parameters>)

pre [<constraint name>]: <Boolean OCL expression>

Q 4(c)

[5 Marks]

Write a postcondition for each of two functions that you have identified in any of your classes.

context <classifier>::<operation> (<parameters>)

post [<constraint name>]: <Boolean OCL expression>

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 20]

Q 5(a)

[5 Marks]

Explain your main role in the Group Project.

Q 5(b)

[5 Marks]

What were the major difficulties faced by your team? And separately by yourself?

Q 5(c)

[5 Marks]

How would you make your finished system more ROBUST, FLEXIBLE, REUSABLE and EFFICIENT?

Q 5(d)

[5 Marks]

If you were now to be given the task of developing the other game, (i.e. either CHESS or BRIDGE) what frameworks could you re-use and how?

[End of Question 5]

Appendix

A software system is being developed to support the operations of a specialised car-hire company: VIP Cars. One software class within the system is called *Reservation* and the lifecycle of a Reservation object is described as follows [based loosely on O'Docherty, Object-Oriented Analysis & Design, Wiley]

“When a Member reserves a CarModel over the Internet, the Reservation is initially *Waiting to be processed* by an Assistant (this is so the Customer can make a Reservation without the intervention of an Assistant). The Reservation becomes *Notifiable* if, some time later, an Assistant finds a suitable unreserved Car in the display area of the car park, or if one is returned by a Customer. In this case, the Car is moved to the reserved area. If no car becomes available for a particular Reservation within a week, the Reservation becomes *NeedingRenewal*: the Member must be contacted, by phone, SMS, email or in person, so that they can cancel the Reservation, or ask for it to be renewed for another week. If the Member cancels or can't be contacted within five days, the Reservation is concluded.

Once a Reservation is Notifiable, the Member must be notified by an Assistant, in Person, by email, SMS or by phone, within three days; if the Customer can be reached, the Reservation is Collectable otherwise it becomes Displayable (a Car that was moved to the reserved area must be returned to the display area). Once a Reservation is Collectable, the Member must collect the Car within three days; if they do collect, the Reservation is Concluded; otherwise, the Reservation becomes Displayable. Once a Displayable reservation's Car has been put back in the display area, the Reservation is Concluded.

At any time, the Member may cancel the Reservation by email, by phone or in person.”

[END OF APPENDIX]

[END OF EXAM]